

Training Configuration and Summary:

Dataset used: rssi_odom_dataset-ORIGINAL.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-07 13:59:35

Training Start: 2025-05-07 13:30:06

Training End: 2025-05-07 13:52:14

Training Duration: 22.15 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 1.8091

MSE: 5.7300

RMSE: 2.3937

R2 Score: 0.0229

Multi-Step Forecast:

Forecast Steps: 40

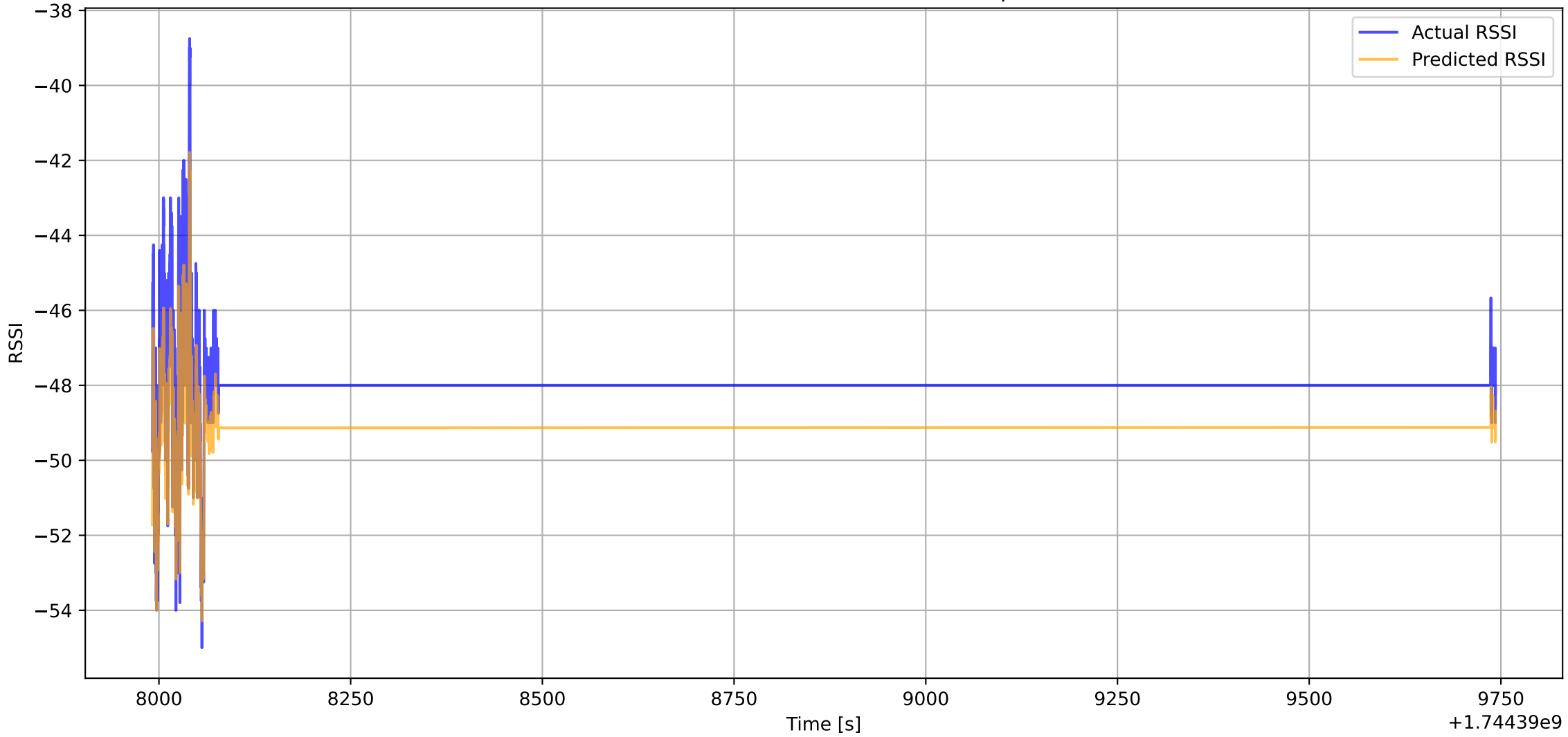
MAE: 0.0210

MSE: 0.0006

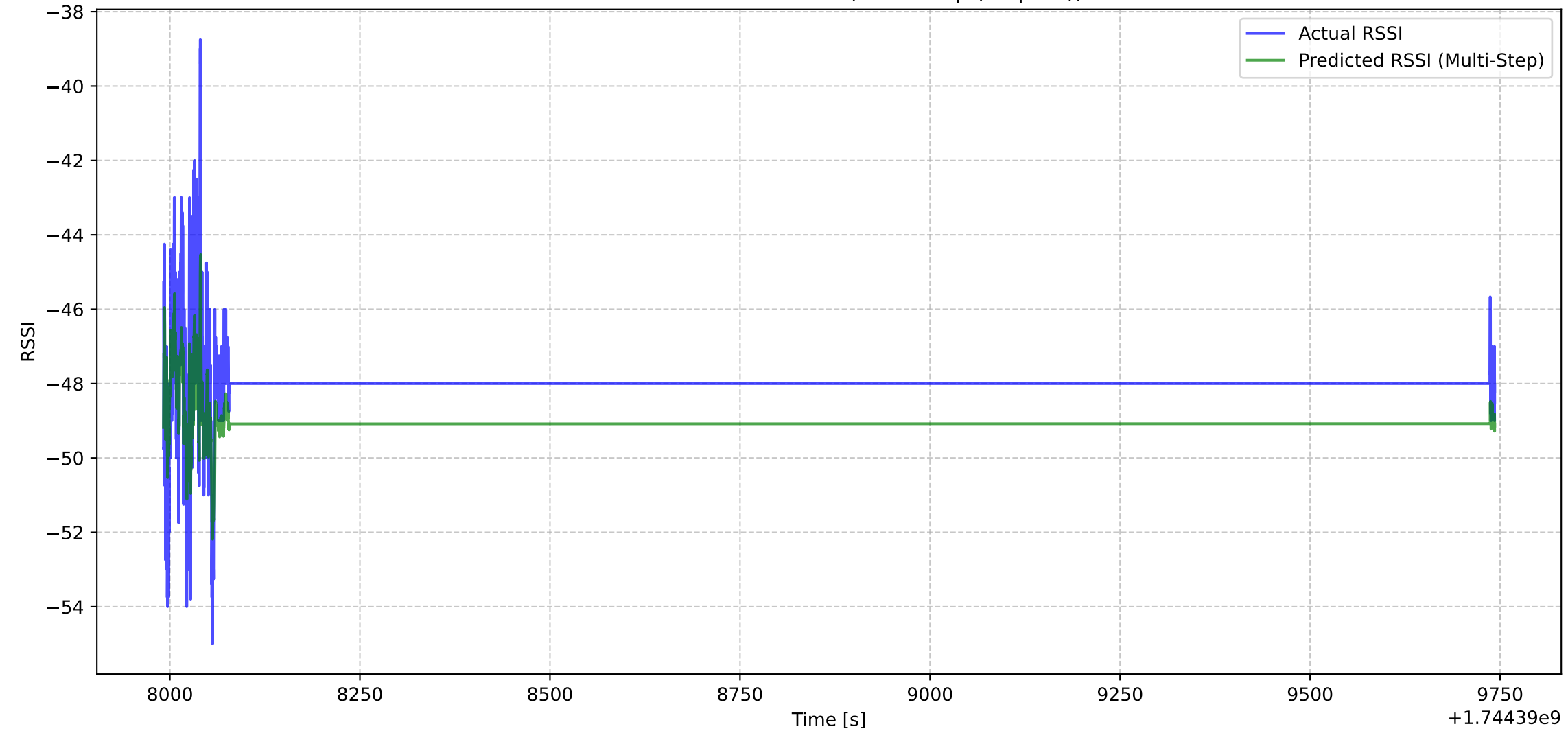
RMSE: 0.0243

R2 Score: 0.5657

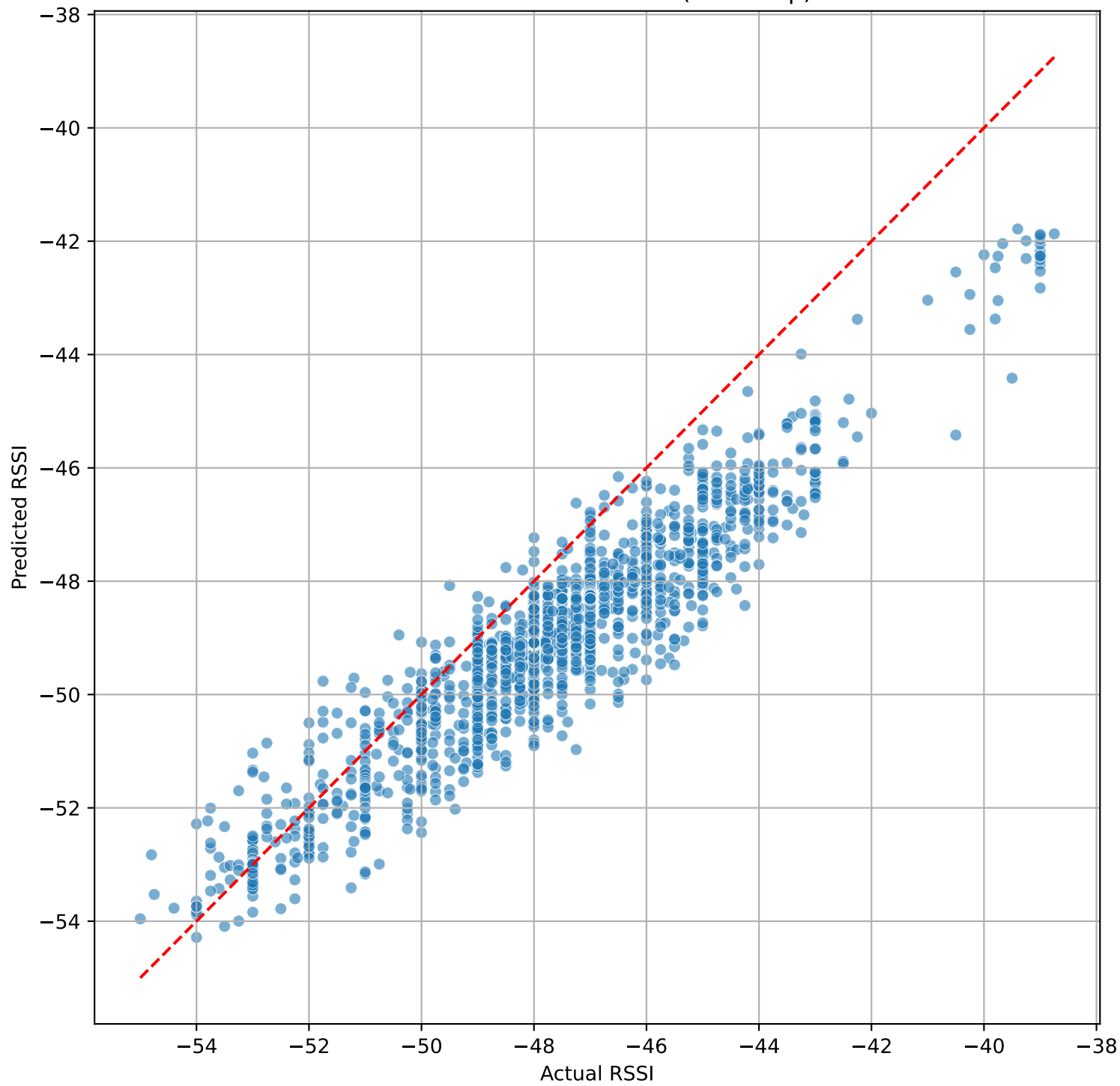
Actual vs Predicted RSSI (First Step)



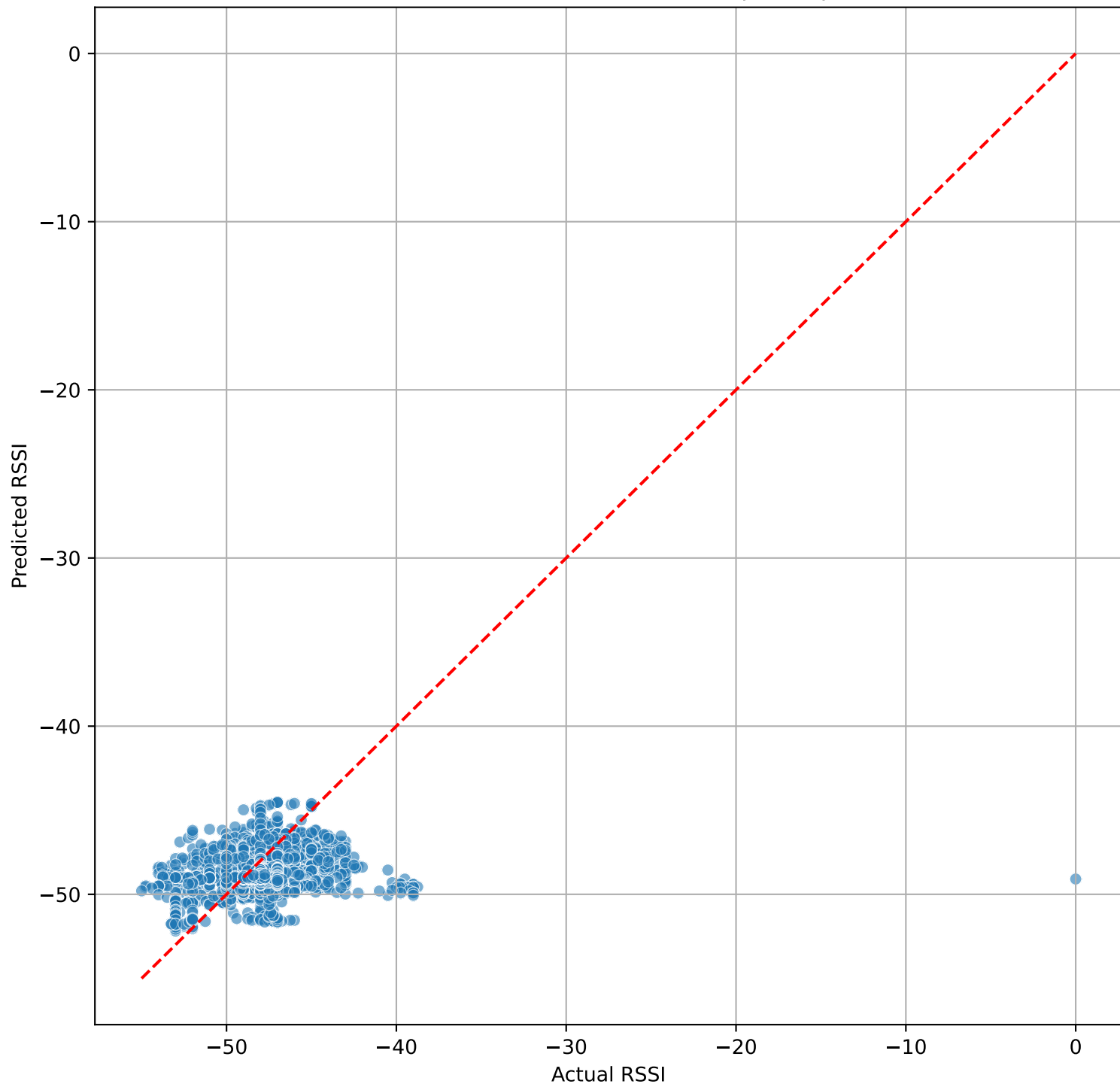
Actual vs Predicted RSSI (Multi-Step (step 40))



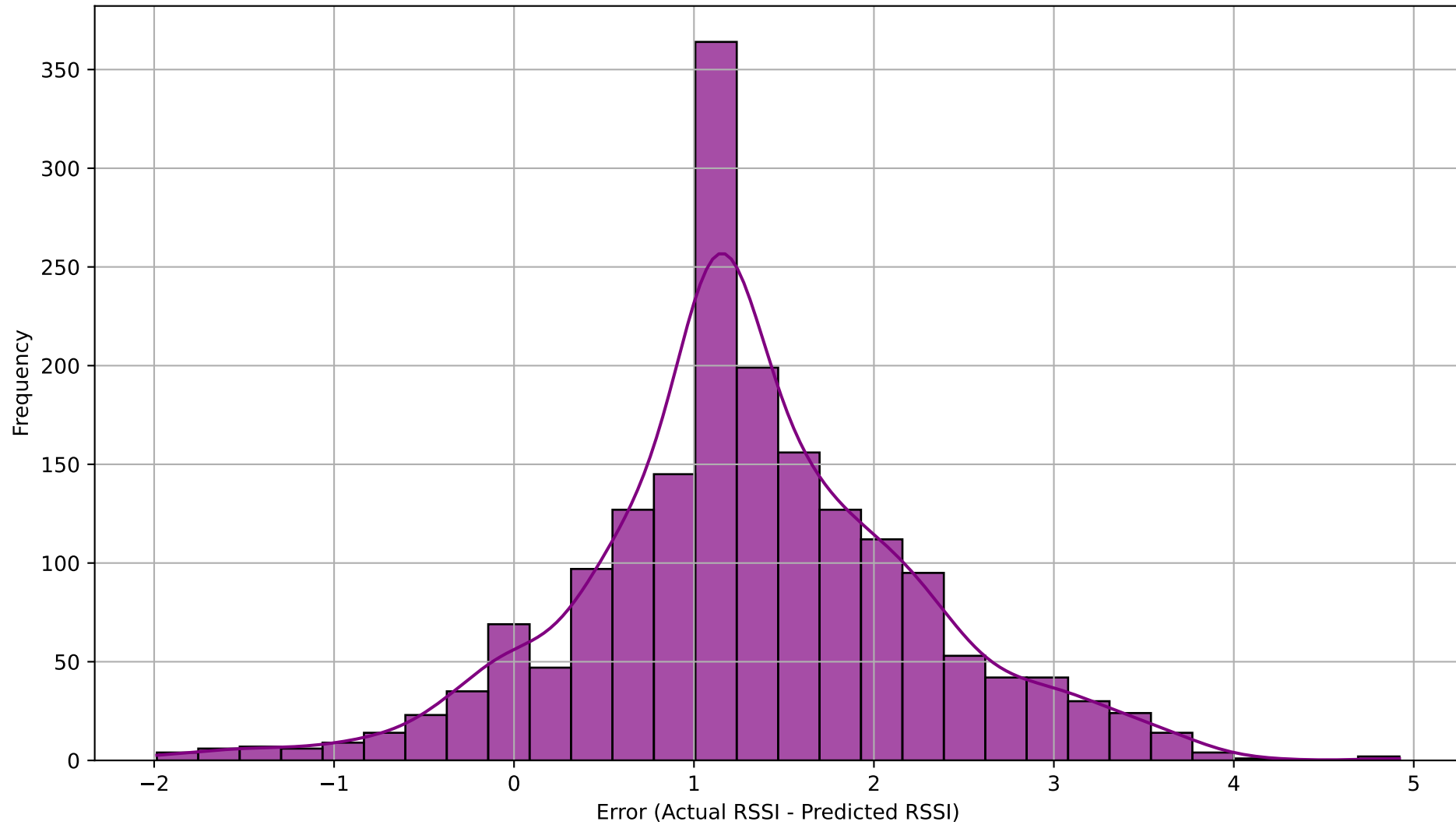
Actual vs Predicted RSSI (First Step)



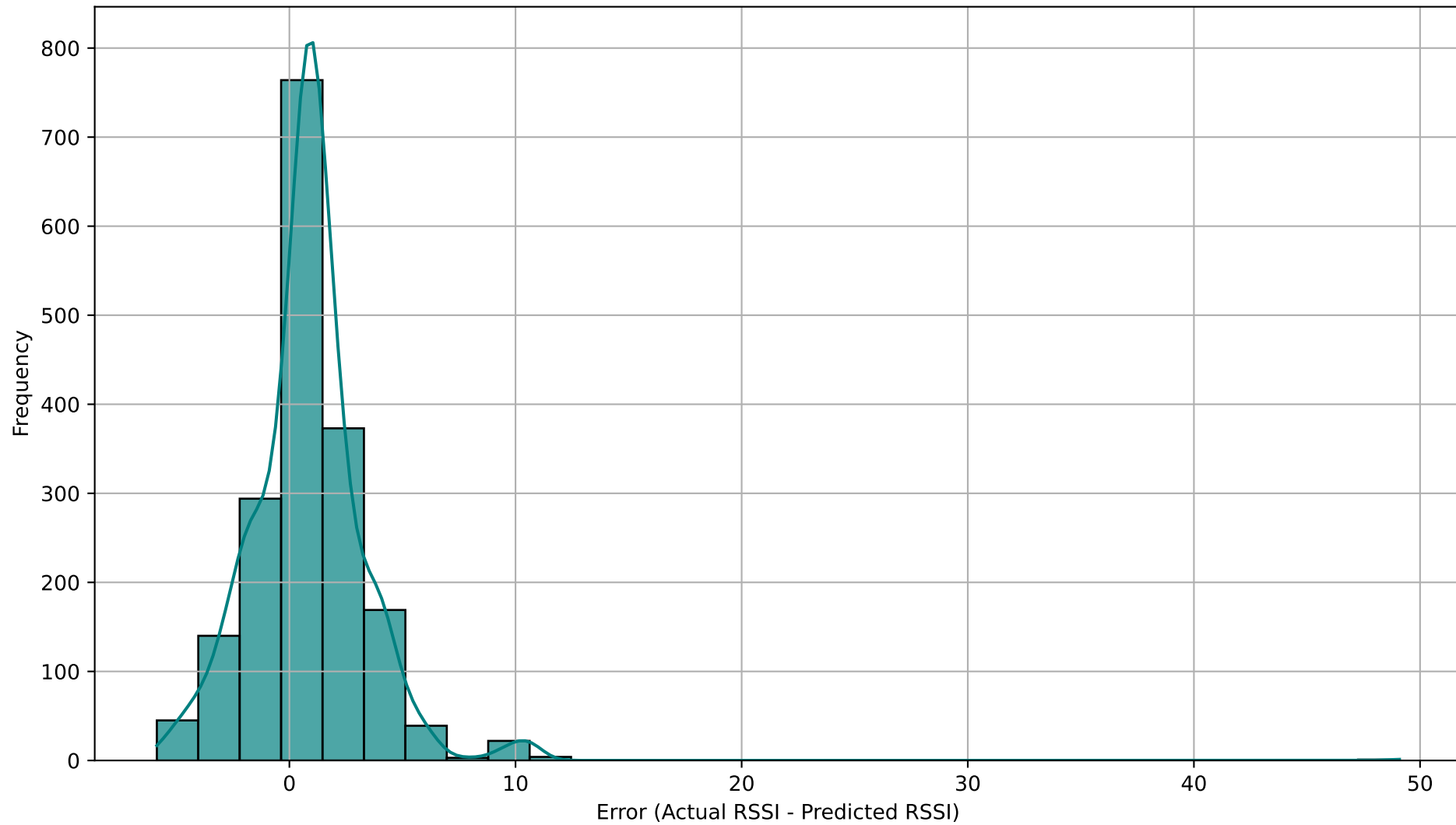
Actual vs Predicted RSSI (Last Step - step 40)



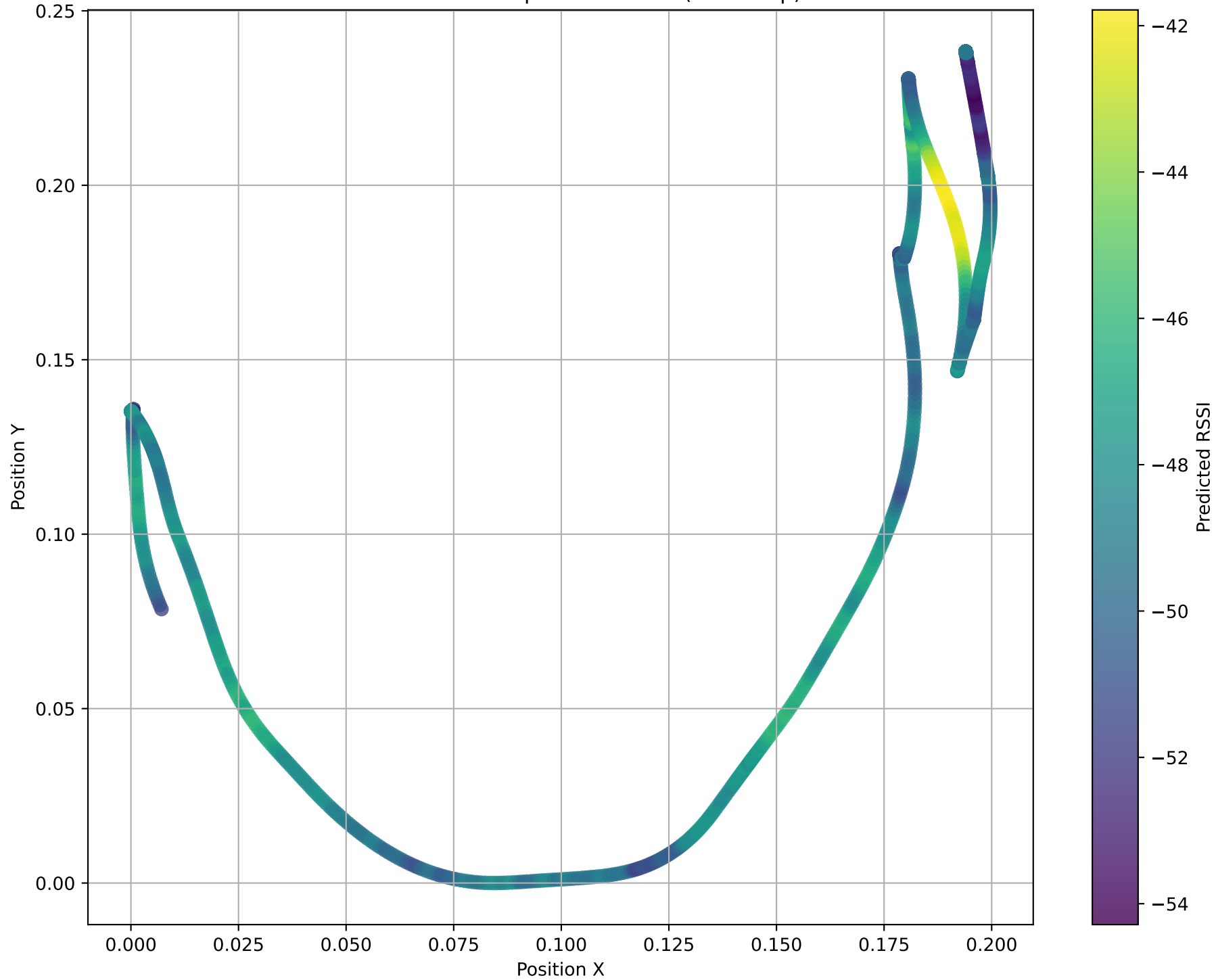
Distribution of Prediction Errors (First Step)



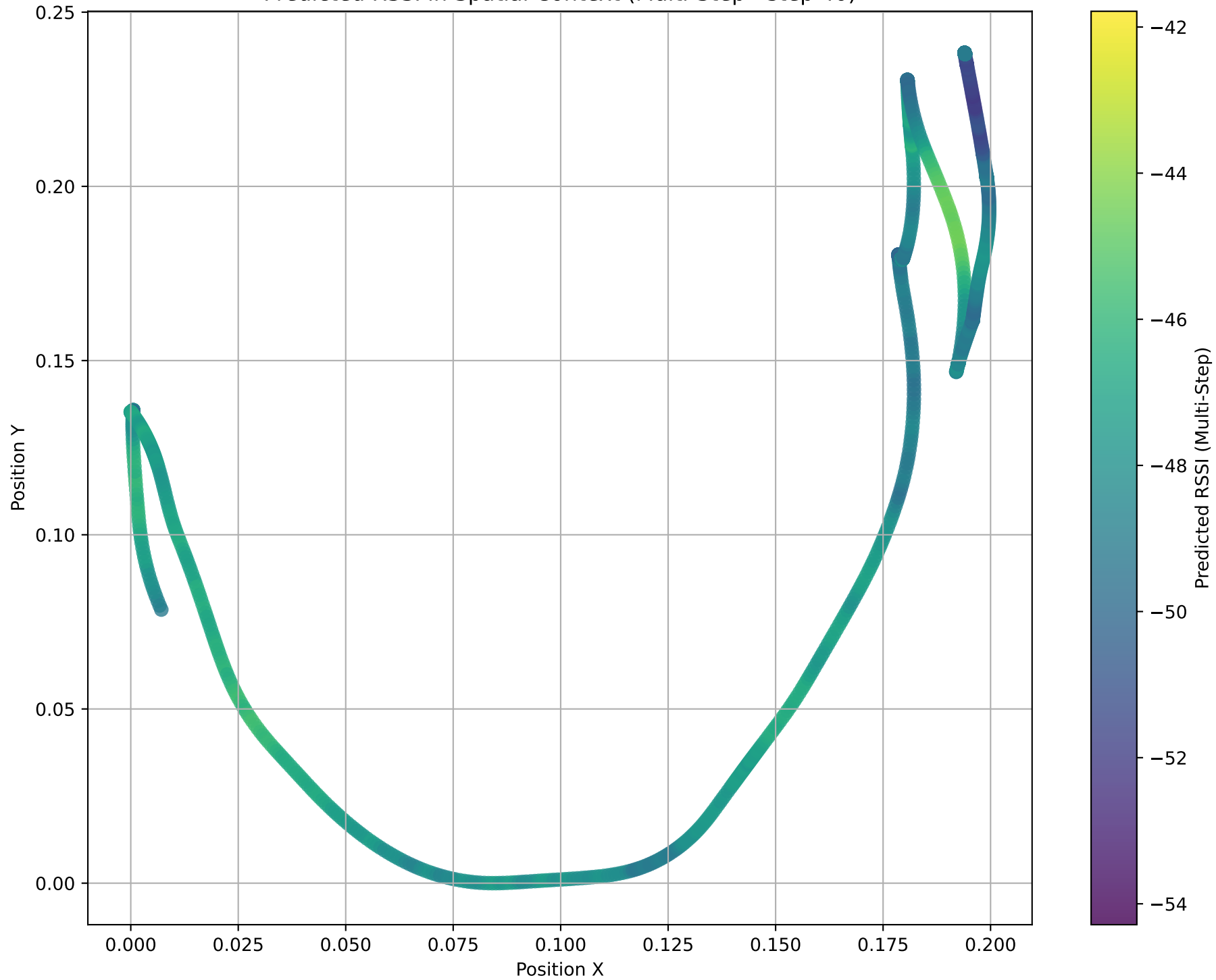
Distribution of Prediction Errors (Last Step - step 40)



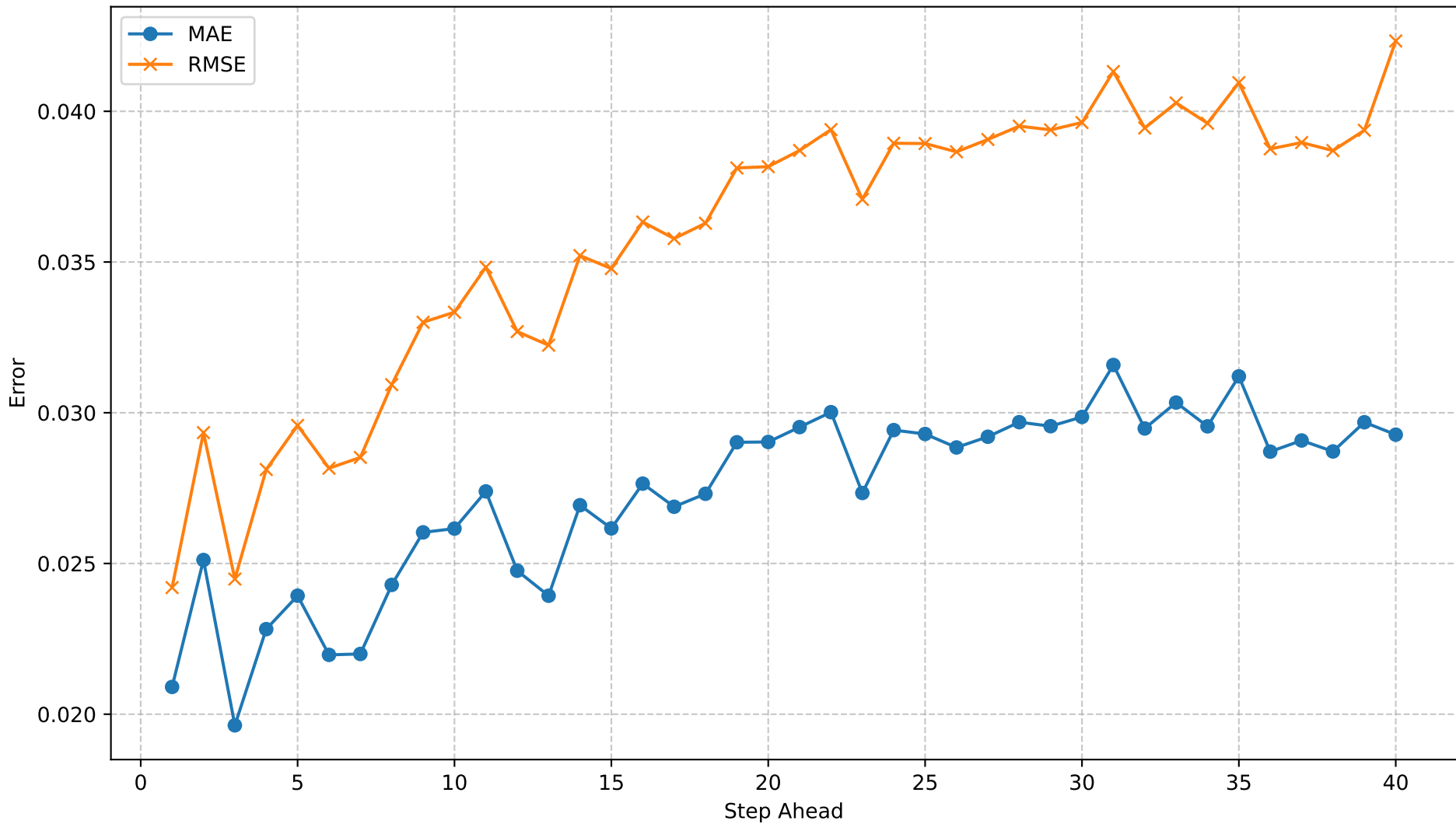
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset-ORIGINAL.parquet

Modelo: LSTM

Forecast steps: 40

MAE / RMSE (dB)

- 1º passo : 0.02 / 0.02
- Mediana : 0.03 / 0.04
- 40º passo : 0.03 / 0.04

Crescimento do erro: 40.0% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 31 (0.03), 35 (0.03), 33 (0.03)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.